

# Hossein Hatamnia

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## Education

### University of California, Riverside (UCR)

Ph.D. in Astronomy

- Advisors: Bahram Mobasher & Drew Newman

Riverside, California, US

Sep 2024 - Present (Expected 2030)

### University of California, Riverside (UCR)

M.S. in Astronomy (en route)

- GPA: 3.98/4

Riverside, California, US

Mar 2026

### Sharif University of Technology (SUT)

B.Sc. in Physics

- GPA: 18.56/20

Tehran, Iran

Sept 2019-Feb 2024

## Research Interests

I study how the cosmic web environment drives galaxy formation and evolution, tracing the interplay between large-scale structure and galaxy properties out to high redshift. I also work on applying AI methods to astronomical data analysis and classification.

## Achievements & Fellowships

- 2026** UCR-Carnegie Observatory 3-Year **Fellowship** (highly competitive graduate fellowship supporting Ph.D. research in collaboration with Carnegie Observatories. Only one student each year.)
- 2024** Dean's **Fellowship**, University of California, Riverside (graduate recruitment fellowship)
- 2026** The Junior Graduate Award (departmental award recognizing outstanding graduate achievement)
- 2024** Institute for Research in Fundamental Sciences (IPM) **Fellowship**
- 2018** **Silver Medal**, Iran's National Astronomy and Astrophysics Olympiad
- 2019** **Top 1% Nationwide**, Iranian National University Entrance Examination

## Publications (12 Papers & 217 Citations)

- 1. Hatamnia, H.**, Mobasher, B., Taamoli, S., Kartaltepe, J. S., Casey, C. M., et al. (2026). Large-scale Structure in COSMOS-Web: Tracing Galaxy Evolution in the Cosmic Web up to  $z \sim 7$  with the Largest JWST Survey. *The Astrophysical Journal*, **1002**(2), Article 192. [doi:10.3847/1538-4357/ae5bac](https://doi.org/10.3847/1538-4357/ae5bac) **Citations: 3**
- 2. Scognamiglio, D.**, Leroy, G., Harvey, D., Massey, R., Rhodes, J., **Hatamnia, H.**, et al. (2026). An ultra-high-resolution map of (dark) matter. *Nature Astronomy*. [doi:10.1038/s41550-025-02763-9](https://doi.org/10.1038/s41550-025-02763-9) **Citations: 3**
- 3. Tanaka, T. S.**, Akins, H. B., Harikane, Y., Silverman, J. D., Casey, C. M., **Hatamnia, H.**, et al. (2025). Discovery of a Little Red Dot Candidate at  $z \gtrsim 10$  in COSMOS-Web Based on MIRI-NIRCam Selection. *The Astrophysical Journal*, 995(1), Article 21. [doi:10.3847/1538-4357/ae145f](https://doi.org/10.3847/1538-4357/ae145f) **Citations: 24**
- 4. Shuntov, M.**, Akins, H. B., Paquereau, L., Casey, C. M., Ilbert, O., **Hatamnia, H.**, et al. (2025). COSMOS2025: The COSMOS-Web galaxy catalog of photometry, morphology, redshifts, and physical parameters from JWST, HST, and ground-based imaging. *Astronomy & Astrophysics*, 704, A339. [doi:10.1051/0004-6361/202555799](https://doi.org/10.1051/0004-6361/202555799) **Citations: 81**
- 5. Khostovan, A. A.**, Kartaltepe, J. S., Salvato, M., Ilbert, O., Casey, C. M., **Hatamnia, H.**, et al. (2026). COSMOS Spectroscopic Redshift Compilation (First Data Release): 488,000 Redshifts Encompassing Two Decades of Spectroscopy. *The Astrophysical Journal Supplement Series*, 282(1), 6. [doi:10.3847/1538-4365/ae1cb9](https://doi.org/10.3847/1538-4365/ae1cb9) **Citations: 65**

6. Khoshtinat, M., **Hatamnia, H.**, & Baghram, S. (2024).

One-point statistics in various cosmic environments in the presence of massive neutrinos.

*Monthly Notices of the Royal Astronomical Society*, 534(2), 1166–1174. doi:10.1093/mnras/stae2140

**Citations: 2**

## Media Coverage

**Hatamnia et al. 2026 — Large-scale Structure in COSMOS-Web (First Author)**

UCR News, RIT news, Space.com, Live Science, ScienceDaily, MSN, Phys.org, (+10 more outlets, May 2026)

**Scognamiglio et al. 2026 — Ultra-High-Resolution Dark Matter Map, *Nature Astronomy***

NASA.gov, National Geographic, CBS News, Scientific American, Smithsonian Magazine, BBC (+10 more outlets, Jan 2026)

**Shuntov et al. 2025 — COSMOS2025 Galaxy Catalog**

Space.com, Discover Magazine, Phys.org, RIT News, UC Santa Barbara News (+10 more outlets, Jun 2025)

## Selected Talks & Poster Presentations

**Large-Scale Structure in COSMOS-Web: Evolution of Galaxies in the Cosmic Web**

Marseille, France

COSMOS Meeting

May 2025

- Contributed Talk

**Radio Spectral Energy Distribution of Star Forming Galaxies at  $z < 1.5$**

Bristol University, Bristol, UK

Towards SKAO (MeerKAT Symposium and Workshop)

Sept 2023

- Poster Presentation

**Radio Spectral Energy Distribution of Star Forming Galaxies at  $z < 1.5$**

Erciyes University, Kayseri, Turkey

IAU/IHOW Radio Astronomy Workshop

Sept 2023

- Contributed Talk

## Research & Teaching Experiences

**Teaching Assistant**

University of California, Riverside

**Undergraduate Physics Laboratories**

Since Winter 2025

**Research Assistant**

Institute of Research in

Fundamental Sciences (IPM)

**Supervisor: Dr. Fatemeh Tabatabaei**

Apr 2024 - Sept 2024

As a member of the MIGHTEE team (PI: Matt Jarvis, Oxford), I use Bayesian MCMC to model radio spectral energy distribution.

**Teaching Assistant**

Sharif University of Technology

**Astrophysics I**

Spring 2023

**Teaching Assistant**

Sharif University of Technology

**Analytical Mechanics I, II**

Sept 2022 - Jul 2023

**Teaching Assistant**

Sharif University of Technology

**Basic programming with Python**

Spring 2021

**Teacher**

High Schools

**Astronomy Olympiad**

Feb 2020 - Apr 2024

## Selected Courses

Data Analysis & High-Performance Computing **Grade: 18.9/20**

Astrophysics I, II **Grade: 18.5, 19/20**

Computational Astrophysics **Grade: 4/4**

Observational Astronomy **Grade: 4/4**

Cosmology **Grade: 4/4**

Gravity and General Relativity **Grade: 18.4/20**

Nonlinear Dynamics and Chaos **Grade: 20/20**

Stellar Structure **Grade: 4/4**

## Skills

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### **Programming**

Python, Julia, C/C++

### **Scientific Libraries**

NumPy, SciPy, Astropy, Matplotlib, OpenCV, LinearAlgebra, emcee

### **Deep Learning**

TensorFlow, PyTorch, Keras, Flux, MLBase

### **Astronomy Software**

EAZY, Bagpipes, DS9, CASA, CARTA, AIPS

### **High-Performance Computing**

Multi-Core Computing, Multi-Threading, CUDA

### **Dev Tools**

Linux, Git, LaTeX